WORKSHEET

Chapter 2: Using De Moivre's theorem

1 Use De Moivre's theorem to simplify each complex number:

a $[\cos(\theta) + i\sin(\theta)]^7$

b $[\cos(\theta) + i\sin(\theta)]^{-3}$

c $[\cos(\theta) - i\sin(\theta)]^5$

d $[\cos(3\theta) + i\sin(3\theta)]^4$

e $[\cos(5\theta) - i\sin(5\theta)]^{-2}$

f $[\cos(4\theta) + i\sin(4\theta)]^8$

2 Evaluate z^4 in Cartesian form if

a $z = 3\left[\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right]$

b $z = 2\left[\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right]$

c $z = \frac{5}{\sqrt{2}}\left[\cos\left(\frac{3\pi}{8}\right) + i\sin\left(\frac{3\pi}{8}\right)\right]$

d $z = 7\left[\cos\left(\frac{5\pi}{12}\right) + i\sin\left(\frac{5\pi}{12}\right)\right]$

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e $z = \frac{3}{\sqrt{2}}\left[\cos\left(\frac{5\pi}{6}\right) + i\sin\left(\frac{5\pi}{6}\right)\right]$

f $z = 2\left[\cos\left(\frac{7\pi}{3}\right) + i\sin\left(\frac{7\pi}{3}\right)\right]$

3 By first converting to polar form, evaluate each of the following, giving your answer in Cartesian form.

a $(1 + i)^4$

b $(2 - i)^4$

c $\left(\sqrt{2} + i\right)^6$

d $\left(2\sqrt{3} + i\right)^5$

e $\left(\sqrt{3} + \sqrt{3}\, i\right)^5$

f $\left(\frac{-1 - i}{\sqrt{3}}\right)^{-4}$

4 Evaluate

a $\frac{1}{(3 + 3i)^4}$

b $\frac{1}{\left(1 - i\sqrt{2}\right)^6}$

c $\frac{1}{(2 + 2i)^8}$

d $\frac{1}{\left(2 - i\sqrt{3}\right)^4}$

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e $\dfrac{1}{(2-2i)^5}$

f $\dfrac{1}{\left(1-i3\sqrt{2}\right)^3}$

5 Simplify each complex number:

a $[\cos(\theta) + i\sin(\theta)]^7 \times [\cos(\theta) + i\sin(\theta)]^{-4}$

b $[\cos(\omega) + i\sin(\omega)]^5 \times [\cos(\omega) + i\sin(\omega)]^8$

c $[\cos(2\beta) + i\sin(2\beta)]^{-2} \times [\cos(2\beta) + i\sin(2\beta)]^6$

d $\dfrac{[\cos(6\alpha) + i\sin(6\alpha)]^9}{[\cos(4\alpha) + i\sin(4\alpha)]^5}$

e $\dfrac{[\cos(5\phi) + i\sin(5\phi)]^7}{[\cos(4\phi) + i\sin(4\phi)]^3}$

f $\dfrac{[\cos(x) - i\sin(x)]^3 \times [\cos(2x) + i\sin(2x)]^8}{[\cos(4x) + i\sin(4x)]^3}$

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Answers

1 **a** $\cos(7\theta) + i\sin(7\theta)$

b $\cos(-3\theta) + i\sin(-3\theta)$

c $\cos(5\theta) - i\,5\sin(\theta)$

d $\cos(12\theta) + i\sin(12\theta)$

e $\cos(-10\theta) - i\sin(-10\theta)$

f $\cos(32\theta) + i\sin(32\theta)$

2 **a** -81

b $-8 + 8\sqrt{3}$

c $\frac{625}{4}$

d $12 - 12\sqrt{3}$

e $\frac{-81}{8} - \frac{81\sqrt{3}}{8}$

f $-8 - 8\sqrt{3}$

3 **a** -4

b $-7 - 24i$

c $-23 - i10\sqrt{2}$

d $58\sqrt{3} + 601i$

e $-36\sqrt{3} - i36\sqrt{3}$

f $-\frac{9}{4}$

4 **a** $-\frac{1}{324}$

b $\frac{23}{729} - \frac{10\sqrt{2}}{729}i$

c $\frac{1}{4096}$

d $\frac{-47}{2401} + \frac{8\sqrt{3}}{2401}i$

e $\frac{-1}{256} - \frac{1}{256}i$

f $\frac{-53}{6859} - \frac{45\sqrt{2}}{6859}i$

5 **a** $\cos(7\theta) + i\sin(7\theta)$

b $\cos(13\omega) + i\sin(13\omega)$

c $\cos(4\beta) + i\sin(8\beta)$

d $\cos(34\alpha) + i\sin(34\alpha)$

e $\cos(23\phi) + i\sin(23\phi)$

f $\cos(4x) + i\sin(4x)$

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